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from machine import Pin, time_pulse_us
import network
import urequests
import time

# WiFi Credentials
SSID = "Wokwi-GUEST"
PASSWORD = ""

# ThingSpeak API Key
API_KEY = "DQQB5CF CBMK8THOT"

# GPIO Pins
TRIG = Pin(3, Pin.OUT)
ECHO = Pin(2, Pin.IN)

LED = Pin(15, Pin.OUT)
BUZZER = Pin(14, Pin.OUT)

# Bin Height (cm)
BIN_HEIGHT = 30

# Connect WiFi
wifi = network.WLAN(network.STA_IF)
wifi.active(True)

print("Connecting to WiFi...")

wifi.connect(SSID, PASSWORD)

while not wifi.isconnected():
    time.sleep(1)
    print("Connecting...")

print("Connected!")
print(wifi.ifconfig())

# Measure Distance
def get_distance():
    TRIG.low()
    time.sleep_us(2)
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TRIG.high()
time.sleep_us(10)

TRIG.low()

duration = time_pulse_us(ECHO, 1, 30000)

if duration < 0:
    return None

distance = (duration * 0.0343) / 2

return distance

# Main Loop
while True:

    distance = get_distance()

    if distance is None:
        print("Sensor Error")
        time.sleep(2)
        continue

    fill = ((BIN_HEIGHT - distance) / BIN_HEIGHT) * 100

    if fill < 0:
        fill = 0

    if fill > 100:
        fill = 100

    print("-----")
    print("Distance :", round(distance, 2), "cm")
    print("Fill Level :", round(fill, 2), "%")

    if fill >= 80:
        status = "FULL"
        LED.on()
        BUZZER.on()
    else:

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        status = "NOT FULL"
        LED.off()
        BUZZER.off()

    print("Status :", status)

    url =
    "https://api.thingspeak.com/update?api_key={}&field1={:.2f}
    {}&field2={:.2f}&field3={}".format(
        API_KEY,
        distance,
        fill,
        1 if status == "FULL" else 0
    )

    try:
        response = urequests.get(url)
        print("ThingSpeak Response:", response.text)
        response.close()
    except:
        print("Failed to upload")

    time.sleep(20)

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